

What Does the Image Add? Text-Conditioned Localization of Complementary Visual Information

CSCI-GA 2271 Computer Vision

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Format: Research Project

Abstract

Given an image and caption, can regional scores identify visible facts that the caption leaves uncovered? The initial 350-image Flickr30K pilot does not support our lightweight scorer’s superiority: its three-seed mean intervention Accuracy@1 is 77.22%, versus 84.50% for inverse crop–text similarity. We will validate labels, correct evaluation ambiguities, and test frozen-model transfer before further model development. A rigorous comparison and failure analysis is a valid course outcome. Publication expansion is conditional on new evidence; if the task is unreliable or adequately solved by strong simple baselines, we will recommend changing direction.

1 Problem, Significance, and Research Questions

Contrastive vision-language pretraining aligns matched images and text in a shared representation space [1]. Existing interpretation methods can localize image patches that contribute to a CLIP representation or matching score [2, 5], while recent masking methods identify concept regions that are important for a prediction [6]. However, *importance* and *complementarity* are not the same. A dog region may be crucial for matching the caption “a man with a dog” but provide little information beyond the text because the dog is already mentioned. Conversely, an unmentioned fire truck may contribute little to the current matching score while containing substantial visual information not stated in the caption.

This distinction matters because natural captions are incomplete by design. Prior work has shown that CLIP does not naturally distinguish captions that complement an image from descriptions that replace it [3], and CIEA shows that preserving image information beyond paired text can improve multimodal retrieval [4]. Yet these works do not directly provide a spatial answer to the question “where is the information the text has not covered?”

We therefore study three questions: **RQ1:** Can we define and learn a region-level score $C(r_k | T)$ that estimates how much semantic content in region r_k remains uncovered by paired text T ? **RQ2:** Does this score respond correctly to controlled changes in the text, decreasing specifically when the missing visual detail is added? **RQ3:** Does the resulting complementarity map generalize across datasets and vision-language encoders, and does it capture a different explanatory signal from ordinary saliency or region importance?

Novelty hypothesis. Complementarity, spatial interpretation and text interventions have substantial precedent. An MLP combining them is insufficient evidence of publication-level novelty. A future contribution must establish valid semantic-coverage labels, a meaningful failure of strong direct baselines, and a useful downstream consequence. An updated comparison

with grounded caption coverage and region-acquisition methods is required before claiming novelty.

2 Related Work

CLIP provides shared image–text representations [1]. Decomposition, Grad-ECLIP and CCI localize representation or decision evidence [2, 5, 6]. They motivate spatial baselines but do not alone establish coverage of a caption’s missing facts.

CIEA extracts complementary retrieval features [4]; caption/description work tests limits of global similarity [3]. CPI measures phrase sensitivity for data selection [7], and Visual Credit Audit tests image contribution beyond text [8]. Our hypothesis needs an explicit comparison with these and grounded coverage methods, rather than treating their combination as established novelty.

3 Method

3.1 Task Definition and Regions

Given image I , paired text T , and candidate visual regions $\{r_k\}_{k=1}^K$, we seek a score

$$q_k = C_\theta(r_k | T) \in [0, 1], \quad (1)$$

where a high value means that region r_k contains visually grounded semantics that are not adequately covered by T . We use *complementarity* as an operational, intervention-defined quantity rather than claiming an exact information-theoretic mutual-information decomposition. In the controlled benchmark, regions come from ground-truth Flickr30K Entities boxes. We will later test model-generated regions from CCI-style patch clustering or GroundedSAM to remove dependence on annotations.

A frozen vision-language encoder produces a region feature v_k and text feature t . The historical model, now treated as a comparator, is a small scorer

$$q_k = g_\theta([v_k, t, v_k \odot t, |P_v v_k - P_t t|]), \quad (2)$$

where P_v and P_t are learned projections when the feature dimensions differ. The scorer is intentionally lightweight so that the research focus remains on the interpretability objective rather than large-scale model training.

3.2 Controlled Text Intervention

Flickr30K Entities links caption phrases to image boxes, allowing us to construct intervention pairs for a target region r_k and phrase p_k [9]. Let T^+ explicitly cover p_k , while T^- omits or generalizes that detail. The core constraint is

$$q_k(I, T^-) > q_k(I, T^+). \quad (3)$$

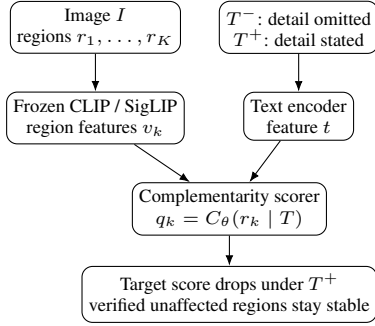


Figure 1: Core idea: complementarity is text-conditioned and must change locally under a controlled caption intervention.

We train with a margin-ranking loss

$$\mathcal{L}_{rank} = \max(0, m - q_k(I, T^-) + q_k(I, T^+)). \quad (4)$$

Locality is valid only for the verified set U of regions whose relevant facts remain unchanged:

$$\mathcal{L}_{loc} = \frac{1}{|U|} \sum_{j \in U} |q_j(I, T^-) - q_j(I, T^+)|. \quad (5)$$

Omit locality when U is empty. The historical implementation assumed all non-target regions unchanged, which can fail for natural captions. We will add a verified matched-control loss using captions in which a non-target phrase is changed. The full objective is $\mathcal{L} = \mathcal{L}_{rank} + \lambda_{loc}\mathcal{L}_{loc} + \lambda_{ctrl}\mathcal{L}_{ctrl}$.

Intervention generation is a major validity concern. We will use three levels of control: (i) deterministic phrase removal/generalization, (ii) matched edits that preserve length and syntax as closely as possible, and (iii) natural multi-caption pairs for the same image when one caption mentions a grounded entity or attribute and another does not. Agent-assisted candidate edits may be used in the Agent Track, but they will be automatically checked for target coverage and manually audited on a stratified sample; failures will be documented in the Agent Log.

4 Initial Evidence and Validity Gate

The existing pilot uses 200/50/100 train/validation/test images and 1,165 pairs. Natural captions retain 97 test images and 251 pairs. Automated split/hash checks passed; independent semantic annotation is incomplete. Historical Acc@1 results are:

Intervention	Inverse cosine	Learned mean
Deletion	84.50%	77.22%
Length-matched	75.75%	74.86%
Natural caption	58.16%	50.17%

These are means of three seed metrics, not prediction ensembles. Each intervention and SigLIP scorer was independently trained: neither establishes frozen-model transfer. The deletion paired learned-minus-inverse difference is -7.28 percentage points, 95% image-bootstrap CI $[-11.70, -3.33]$. This favors

the baseline at this scale without proving the broader question impossible.

Phrase deletion is a proxy for semantic coverage. A category word may cover identity while omitting an attribute. Define fact granularity and distinguish omitted, covered, contradicted and uncertain facts. Natural captions may change multiple regions; phrase/ID exclusion cannot rule out synonyms. Repeated filler tokens do not ensure syntax-matched edits. Audit 100 stratified pairs for visibility, omission, grammaticality and non-target changes before interpreting comparative scores.

Historical absolute-localization code favors target index zero on ties and needs recomputation. Raw TIG and drift are scale-sensitive. Spatial center-crop features and full region crops may see different target content. Attribution baselines are task adaptations, not complete official reproductions. Preserve these historical measurements with caveats.

The next executable experiment holds deletion-trained checkpoints fixed and evaluates matched/natural captions without fitting. Report per-seed results, true prediction ensembles, tie-aware absolute and difference rankings, normalized changes and paired image-bootstrap intervals. Semantic labels remain human-review dependent. Strong direct region-description plus textual entailment/coverage baselines and an independent reader are required before publication expansion.

5 Experiments and Evaluation

Primary benchmark. Flickr30K Entities contains 31,783 images, 158,915 captions, and 276k manually annotated bounding boxes linked to caption phrases [9]. It is well suited to controlled region-text interventions. We will split intervention templates and images by the official train/validation/test partitions to prevent leakage.

Conditional, unrun generalization benchmark. Visual Genome provides dense region descriptions, boxes, objects, attributes, and relations [10]. We will use it to test whether a scorer trained on Flickr30K Entities transfers to a denser vocabulary and different annotation style.

Baselines. We will compare against (i) random and region-area controls; (ii) inverse region-text cosine similarity; (iii) Gandelsman-style patch contribution; (iv) Grad-ECLIP saliency aggregated over regions; and (v) CIEA-inspired complementary feature score. The key question is whether these existing importance or similarity signals satisfy our intervention-specific tests without being trained for complementarity.

Metrics. Distinguish single-caption ranking from intervention-difference ranking. The latter is a sensitivity diagnostic. A supplementary metric is the *Target Intervention Gap*

$$\text{TIG} = q_k(I, T^-) - q_k(I, T^+), \quad (6)$$

which should be positive for a valid target edit; magnitude depends on score scale. *Off-Target Stability* measures the mean absolute score change over verified unchanged regions and should be small. With ground-truth regions, we report target-region ranking Accuracy@1 and MRR; with free-form masks,

we report pointing accuracy and IoU against the grounded box. We also report deletion faithfulness by masking high-complementarity regions and measuring loss of recoverability for the omitted phrase. Report three seeds, a separately labeled prediction ensemble, paired image-bootstrap intervals, expected ranking under random tie-breaking, candidate-count strata, and scale-normalized gaps. Same-CLIP masked similarity is only a proxy for recoverability. An independent reader is required to measure newly recovered correct facts.

Ablations. We will test ground-truth boxes versus model-generated regions, CLIP versus SigLIP, crop features versus pooled patch features, each loss term, different intervention constructions, and training on one dataset versus transfer to the other. A critical negative control edits an unrelated caption phrase; a valid complementarity method should not move the target region score substantially.

Deferred applications. Finance and other downstream domains are outside the next round. They should be reconsidered only after valid core evidence and a clear scientific reason for expansion.

6 Deliverables, Risks, Verification, and Timeline

Expected deliverables. (1) A reproducible benchmark builder for text-conditioned region interventions; (2) a complementarity scorer and visualization pipeline; (3) implementations of strong spatial-interpretability baselines; (4) quantitative Flickr30K evaluation, with Visual Genome conditional on the validity gate; (5) ablations and qualitative failure analysis; (6) code repository with incremental commits and a research-paper-style final report; and (7) the required Agent Log. The optional finance testbed is an extension rather than a core deliverable.

Decision gates. First establish label validity and frozen-model transfer. If simple baselines remain strongest, finish a rigorous course comparison and failure analysis. If labels cannot define a useful task, or no meaningful residual problem remains, stop the learned-method direction. A possible replacement is budgeted region acquisition: select the next region revealing the most new correct facts beyond an existing caption, using equal budgets and a fixed reader. This requires its own literature and feasibility checks, not an assumed novelty claim. At most two motivated variants (calibrated baseline residual and independently supervised coverage) may follow valid evidence; no unbounded tuning.

Agent Track verification. Agent tools may assist with literature search, boilerplate code, experiment scripting, caption-edit candidates, and report/figure preparation. We will independently verify generated code through unit tests and small hand-checked examples, reproduce key baseline behaviors before extension, audit intervention data, rerun core experiments across seeds, and maintain an Agent Log emphasizing incorrect suggestions, implementation failures, and how each result was verified. Every team member will be prepared to explain any submitted code during the final walkthrough.

Compute and execution. Use any available compatible GPU, including L4 or A100; neither is mandatory. Reconcile the ear-

lier 20-hour authorization and imported 24-hour HPC handoff, deduct final historical usage, and use the lower cap if unresolved. Next diagnostics have a maximum of four additional allocated GPU hours within the remaining balance. Stage code, models and data before allocation; run bootstrap/reporting on CPU. Monitor progress/utilization every five seconds. Target 70–90% during steady GPU-ready work where feasible; diagnose sustained < 30% for two minutes. These are internal targets, not cluster rules. Release idle allocations; never repeat useless work to raise utilization. Subagents may prepare independent work; one owner controls submissions and budget.

Revised sequence. P0: recover artifacts, provenance and final accounting. P1: audit 100 stratified semantic pairs. P2: run frozen deletion-trained checkpoints on matched/natural captions, then CPU evaluation. P3: decide whether to conclude the course comparison, try at most two motivated variants, or change direction. P4: only after a meaningful validation signal, confirm on untouched images and expand to independent regions/readers/datasets. Existing test images are exploratory after repeated inspection. Runnable commands are in `hpc/NEXT_ROUND.md`; publication work is separately gated.

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